

10-403: Deep Reinforcement Learning

Instructors: Aviral Kumar & Katerina Fragkiadaki

Carnegie Mellon University
School of Computer Science

Course Details

- **Course website:** <https://cmudeeprl.github.io/> ,all you need to know
- Please join piazza and gradescope linked on the webpage.
- **Attendance:** In-person preferred, though if you cannot attend some lectures that's OK.
- Unofficially, people can audit the course!
- **Resources:** should be runnable on your laptops, if not, let us know!
- **Grading:**
 - 5 homework assignments: implementation and question/answering: 50%
 - 3 quizzes: 50%
- The readings on the schedule are **required** unless noted otherwise

Course Staff: Instructors



Instructor **Katerina Fragkiadaki**

Email: katef@cs.cmu.edu

Office hours: Monday & Wednesday
10:50 - 11:20 AM

- **Who:** Associate Professor in MLD
- **Research Interests:** Robotics, computer vision, reinforcement learning (RL), diffusion models
- **Preferred contact:** Email



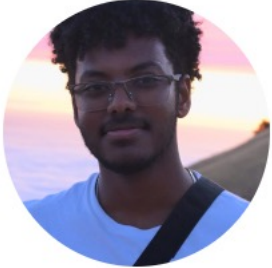
Instructor **Aviral Kumar**

Email: aviralku@andrew.cmu.edu

Office hours: Monday & Wednesday
10:50 - 11:20 AM

- **Who:** Assistant Professor in CSD & MLD. PhD from UC Berkeley, Faculty for 1 semester
- **Research Interests:** Anything and everything reinforcement learning (RL), *will talk about some of it today*
- **Preferred contact:** Email

Course Staff: Teaching Assistants



TA [Eyob Dagnachew](#)

Email: edagnach@andrew.cmu.edu

Office hours: Mondays 4:00 - 5:00pm

Location: GHC 8228



TA [Douglas Chen](#)

Email: dchen3@andrew.cmu.edu

Office hours: Thursday 5:00 - 7:00pm

Location:



TA [Arshika Lalan](#)

Email: alalan@andrew.cmu.edu

Office hours: Tuesday 11:30 - 12:30pm

Location:



TA [Marian Qian](#)

Email: marianq@andrew.cmu.edu

Office hours: Thursdays 3:00 - 4:00pm

Location: GHC 8115



TA [Yingtao Luo](#)

Email: yingtaol@andrew.cmu.edu

Office hours: TBD

Location:

Course Logistics

Prerequisites:

- Prerequisites: 10301/10601 introduction to machine learning
- Minimum Grades: B in 10301/10601
- Corequisites: None
- Anti-requisites: None
- Anti-req Prohibits: None

Prerequisite Knowledge:

Students should have a solid understanding of the following areas

- Algorithms: e.g., What problem does Dijkstra's algorithm solve?
- Probability: e.g., What is Bayes rule? How do you normalize a distribution?
- Computer vision: convolutional networks, object detection architectures, LSTMs, attention models
- Deep Learning: familiarity with TensorFlow and Pytorch.
- Matrix Calculus: e.g., What are derivatives of matrix-matrix and matrix-vector products? What is the multivariate chain rule?
- Programming: e.g., What are classes and inheritance? How do you structure read data from files? How do you plot figures to visualize results?
- Numerical programming: e.g., How would you perform an elementwise product instead of an inner product? How do you invert a matrix?

Course Content and Goals

Goal: Learn about deep reinforcement learning (“deep RL”) techniques and become ready to apply them in practice

- Learn to understand deep RL ideas and how it connects to “non-deep” RL
- Learn to implement these ideas and techniques
- Learn to apply RL onto new application domains

Goal: Learning to Act (Make Decisions)

Building agents that **learn** to act and accomplish **tasks / goals** in **dynamic** environments



...as opposed to agents that execute **pre-programmed** behaviors in **static** environments...



Motion and Action are Important

“The brain evolved, not to think or feel, but to control movement.”

Daniel Wolpert



Daniel Wolpert: The real reason for brains | TED Talk | TED.com

https://www.ted.com/talks/daniel_wolpert_the_real_reason_for_brains ▼

What is Reinforcement Learning?

Mathematical paradigm for learning to act via trial and error

Supervised learning

Given a dataset, find the pattern that helps ***predict*** the label for this input

- Generally, assumes i.i.d. data
- Previous predictions don't affect other predictions.
- Typically, one-step of a problem

Reinforcement learning

Sequential decision-making:

Observe the world → Take an action
→ Observe → Take an action → ...

- Data is ***not*** i.i.d.
- Previous actions determine how “hard” it would be to do in future

Where Does RL Come From?

An old concept and study area in psychology.

“Behavior is primarily shaped by reinforcement rather than free-will.”

- *behaviors that result in praise/pleasure tend to repeat,*
- *behaviors that result in punishment/pain tend to become extinct.*



B.F. Skinner
1904-1990
Harvard psychology

Formalizing RL as Learning from Trial and Error

Learning behaviors that maximize a reward function via interaction

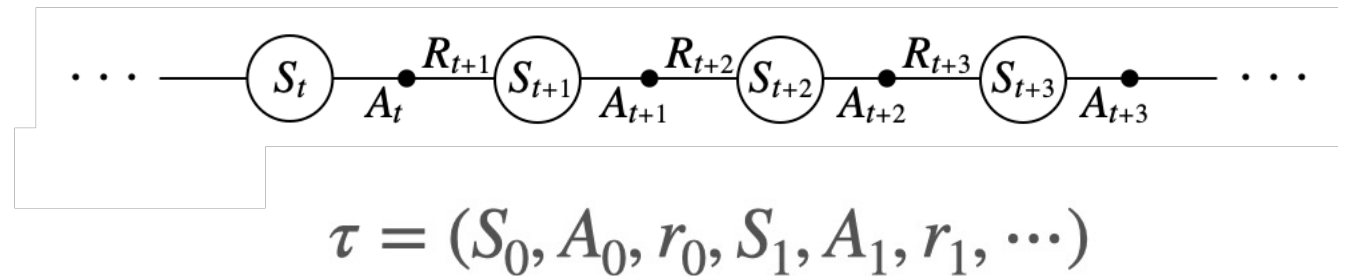
Agent and environment interact at discrete time steps: $t = 0, 1, 2, K$

Agent observes state at step t : $S_t \in \mathcal{S}$

produces action at step t : $A_t \in \mathcal{A}(S_t)$

gets resulting reward: $R_{t+1} \in \mathbb{R}$

and resulting next state: $S_{t+1} \in \mathcal{S}^+$



$$J(\pi) = \max_{\pi} \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^T r_t \right]$$

Why RL? Standard ML Models the Data

Unsupervised learning

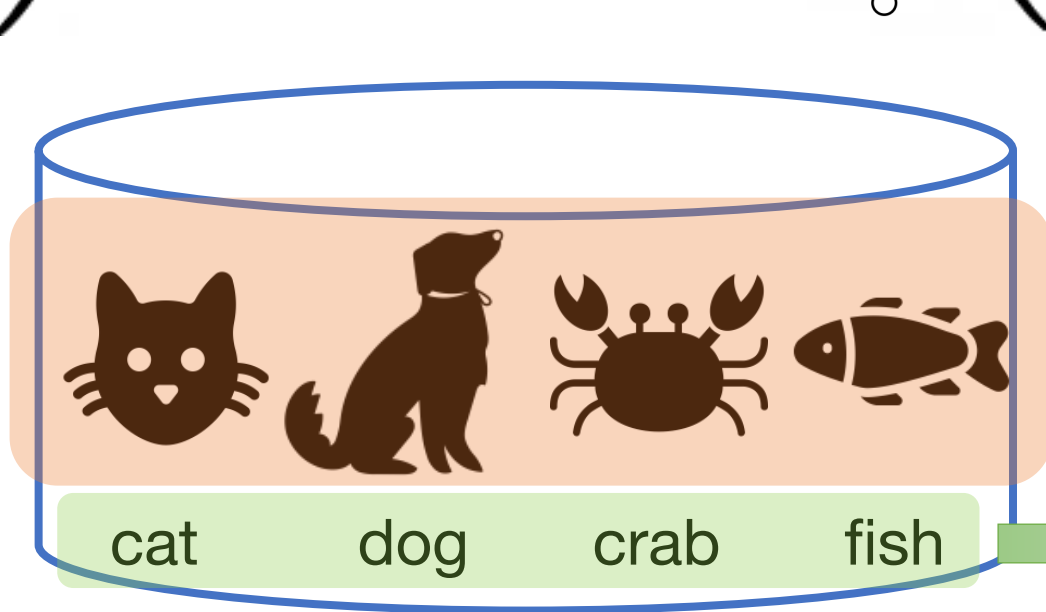
$$p(\text{cat image})$$

Supervised learning

$$p(\text{cat} \mid \text{cat image})$$

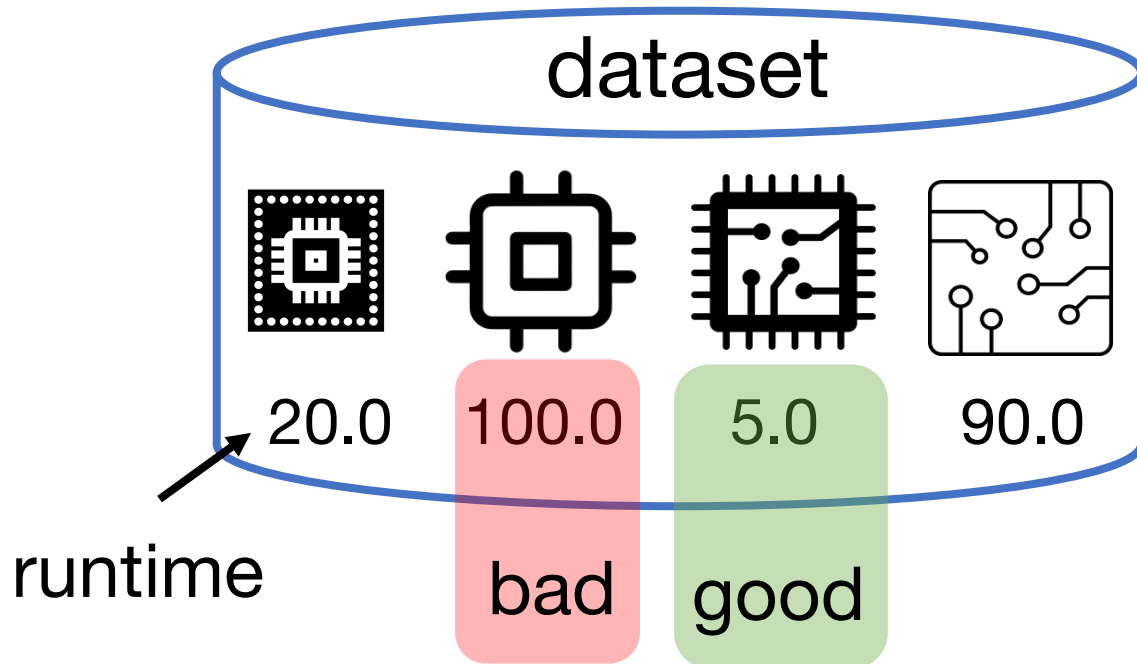
image

label



But Modelling the Data Is Not Enough

Goal: Discover a chip that is better than dataset



Supervised learning

$$\text{Model} \approx \text{Runtime}(\text{Chip})$$

Unsupervised learning

$$\text{New chip} \sim p(\text{Chip})$$

In expectation, chips (**decisions**)
not better than the dataset!

Even Mimicking “High-Quality” Data is Not Enough

Show that the inequality

$$\sum_{i=1}^n \sum_{j=1}^n \sqrt{|x_i - x_j|} \leq \sum_{i=1}^n \sum_{j=1}^n \sqrt{|x_i + x_j|}$$

holds for all real numbers $x_1, x_2, \dots, x_n$.

Analyzing individual terms is not good enough

The model “**asserts**” its way out, no realization of its mistake / backtracking

Analyzing $\sqrt{|x_i - x_j|}$ and $\sqrt{|x_i + x_j|}$

1. Absolute Value Cases:

- When $x_i = x_j$, $\sqrt{|x_i - x_j|} = \sqrt{0} = 0$ and $\sqrt{|x_i + x_j|} = \sqrt{|2x_i|}$.
- When $x_i \neq x_j$, we need to analyze how $|x_i - x_j|$ compares to $|x_i + x_j|$.

2. Comparing $|x_i - x_j|$ and $|x_i + x_j|$:

Consider two cases for (x_i, x_j) :

- **Case 1:** $x_i \geq 0$ and $x_j \geq 0$: Here both x_i and x_j are non-negative. We have:

$$|x_i - x_j| = x_i - x_j \quad \text{and} \quad |x_i + x_j| = x_i + x_j.$$

Clearly, $x_i + x_j \geq x_i - x_j$ if $x_i \geq x_j$ or $x_j \geq x_i$.

- **Case 2:** $x_i \leq 0$ and $x_j \leq 0$: Both are non-positive. We have:

$$|x_i - x_j| = -(x_i - x_j) = x_j - x_i \quad \text{and} \quad |x_i + x_j| = -(x_i + x_j) = -(x_i + x_j).$$

Similarly, $|x_i + x_j|$ would be $|-x_i - x_j|$ and the same argument holds.

- **Case 3:** $x_i \geq 0$ and $x_j \leq 0$ or vice versa: This makes $(x_i - x_j)$ and $(x_i + x_j)$ different in signs or their modulus. The absolute values may change based on these signs but we still have the result as:

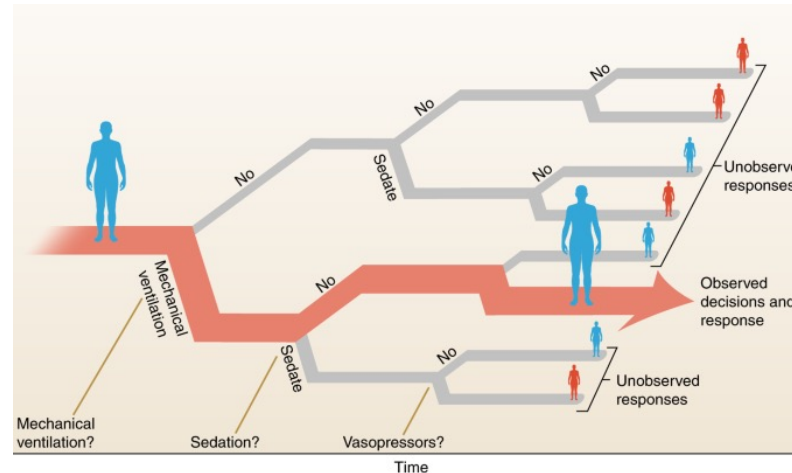
$$\sqrt{|x_i + x_j|} \geq \sqrt{|x_i - x_j|}$$

Same Desideratum for Many Problems

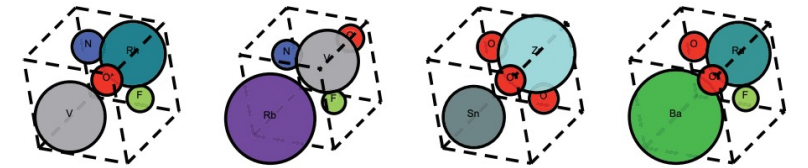
Goal: Discover decisions better than data



Decision: robot policy
Better: task success



Decision: treatment plans
Better: recovery rate



Decision: material structure
Better: Energy

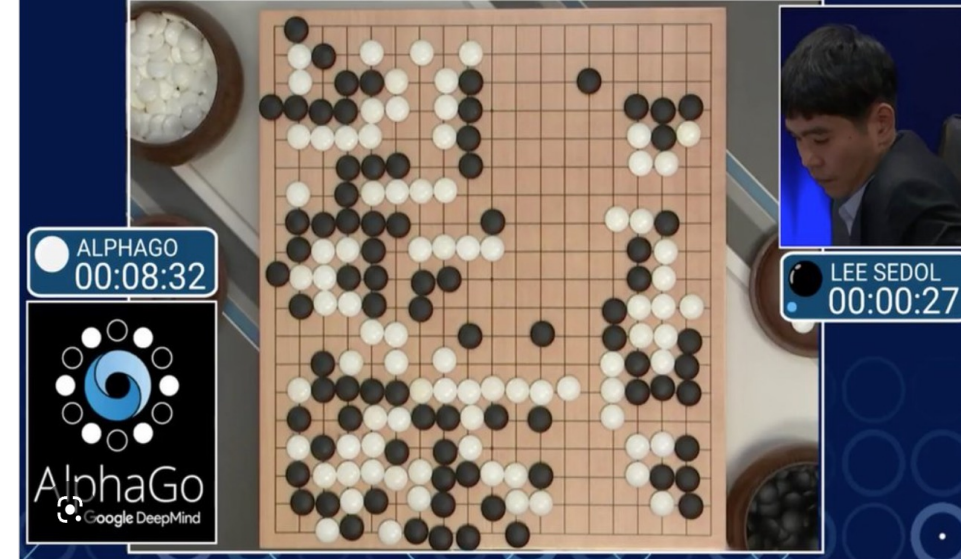
How Can We Discover Better Decisions?

Discover better decision =

Optimize decisions to maximize some utility

This is what RL tries to study

Reinforcement learning (RL)
Bandits
Bayesian Optimization



~millions of games

RL vs Supervised Learning (Formally...)

More formally...

$$\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$$

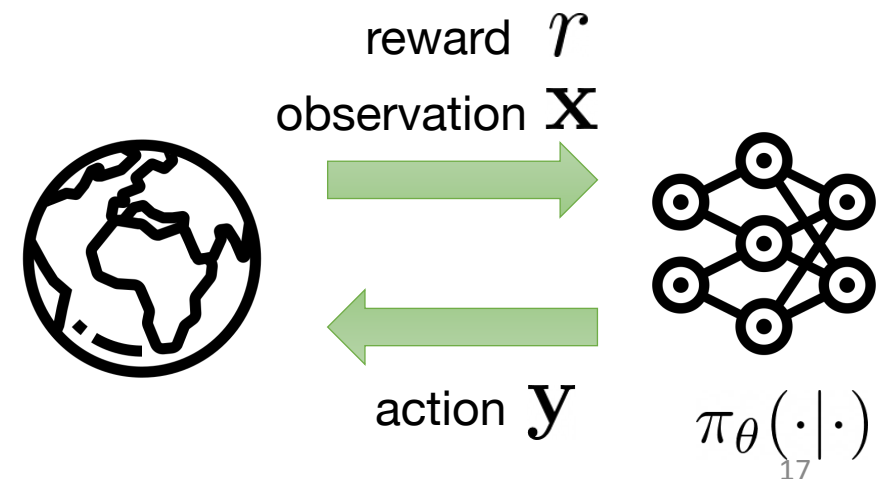
$$\max_{\pi} \sum_{i=1}^n \mathbb{E}_{\mathbf{y}_i \sim \mathcal{D}} [\log \pi(\mathbf{y}_i | \mathbf{x}_i)]$$

Maximize likelihood
of seen data

$$\max_{\pi} \sum_{i=1}^n \mathbb{E}_{\mathbf{y}_i \sim \pi(\cdot | \mathbf{x}_i)} [r(\mathbf{x}_i, \mathbf{y}_i)]$$

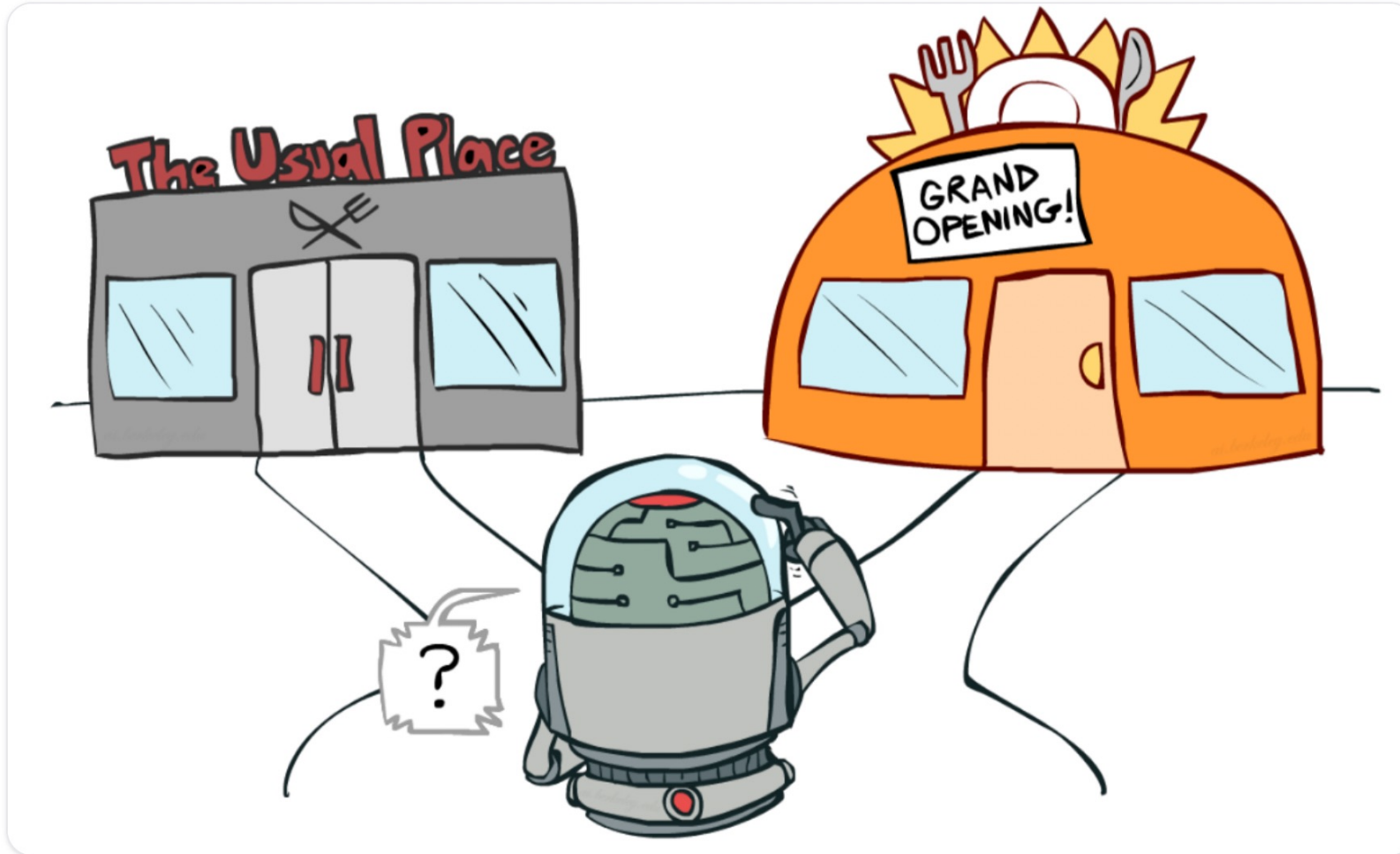
Optimize an
unknown function r

Typically addressed via continuous, **active** interaction

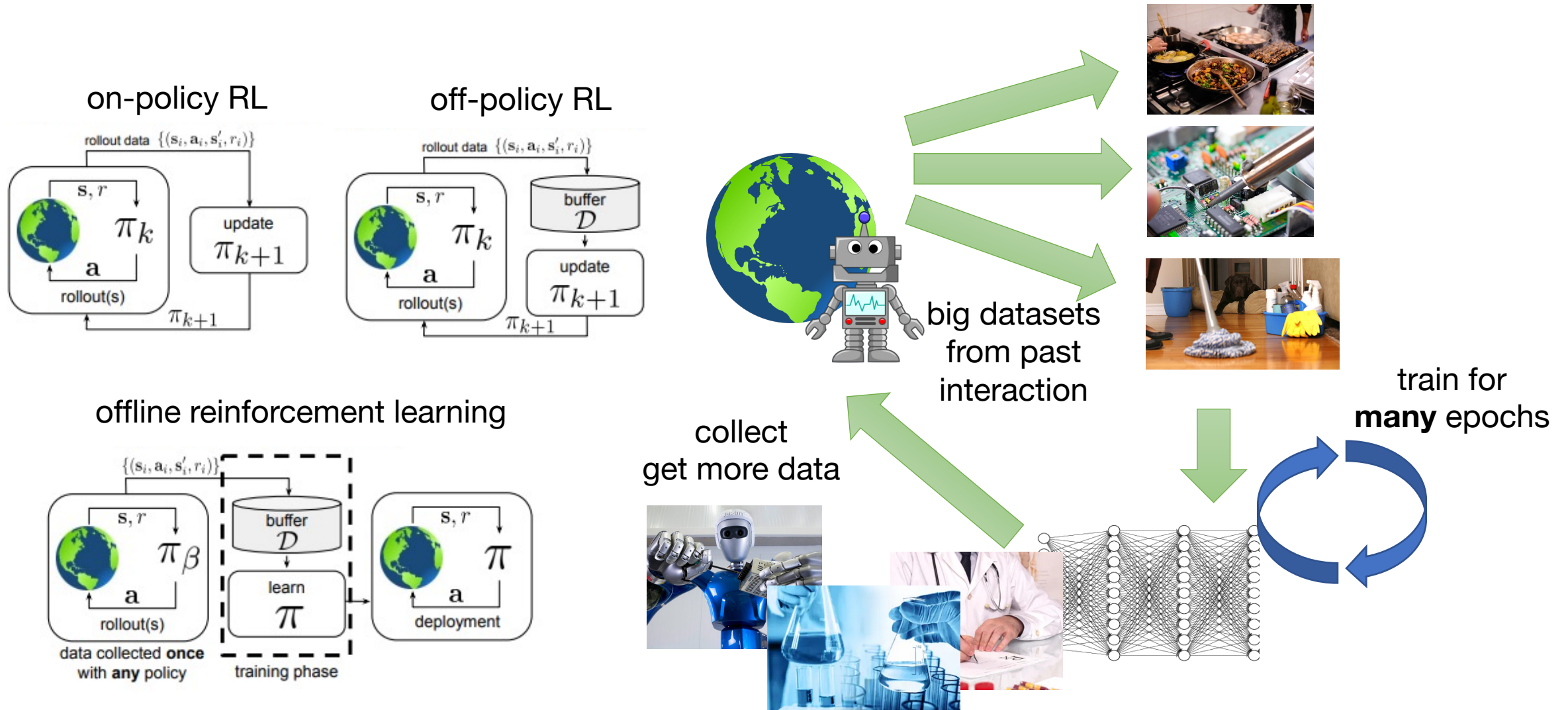


RL vs Supervised Learning (Formally...)

The exploration-exploitation problem



Deep RL Bridges the Gap Between RL & DL



Why is Deep RL Promising in the Long Run?



This is a big lesson. As a field, we still have not thoroughly learned it, as we are continuing to make the same kind of mistakes. To see this, and to effectively resist it, we have to understand the appeal of these mistakes. We have to learn the bitter lesson that building in how we think we think does not work in the long run. The bitter lesson is based on the historical observations that 1) AI researchers have often tried to build knowledge into their agents, 2) this always helps in the short term, and is personally satisfying to the researcher, but 3) in the long run it plateaus and even inhibits further progress, and 4) breakthrough progress eventually arrives by an opposing approach based on scaling computation by search and learning. The eventual success is tinged with bitterness, and often incompletely digested, because it is success over a favored, human-centric approach.

One thing that should be learned from the bitter lesson is the great power of general purpose methods, of methods that continue to scale with increased computation even as the available computation becomes very great. The two methods that seem to scale arbitrarily in this way are *search* and *learning*.

Search / Optimization

Learning

*Summary
So Far...*

Introduction to RL

- **Definition:** A mathematical formalism for making decisions.
- **Formulation:** Maximize **cumulative** reward over interaction with an environment.
- **Comparisons:** Different from supervised learning, which is **static** in nature, doesn't require maximization, and is inherently not **sequential**

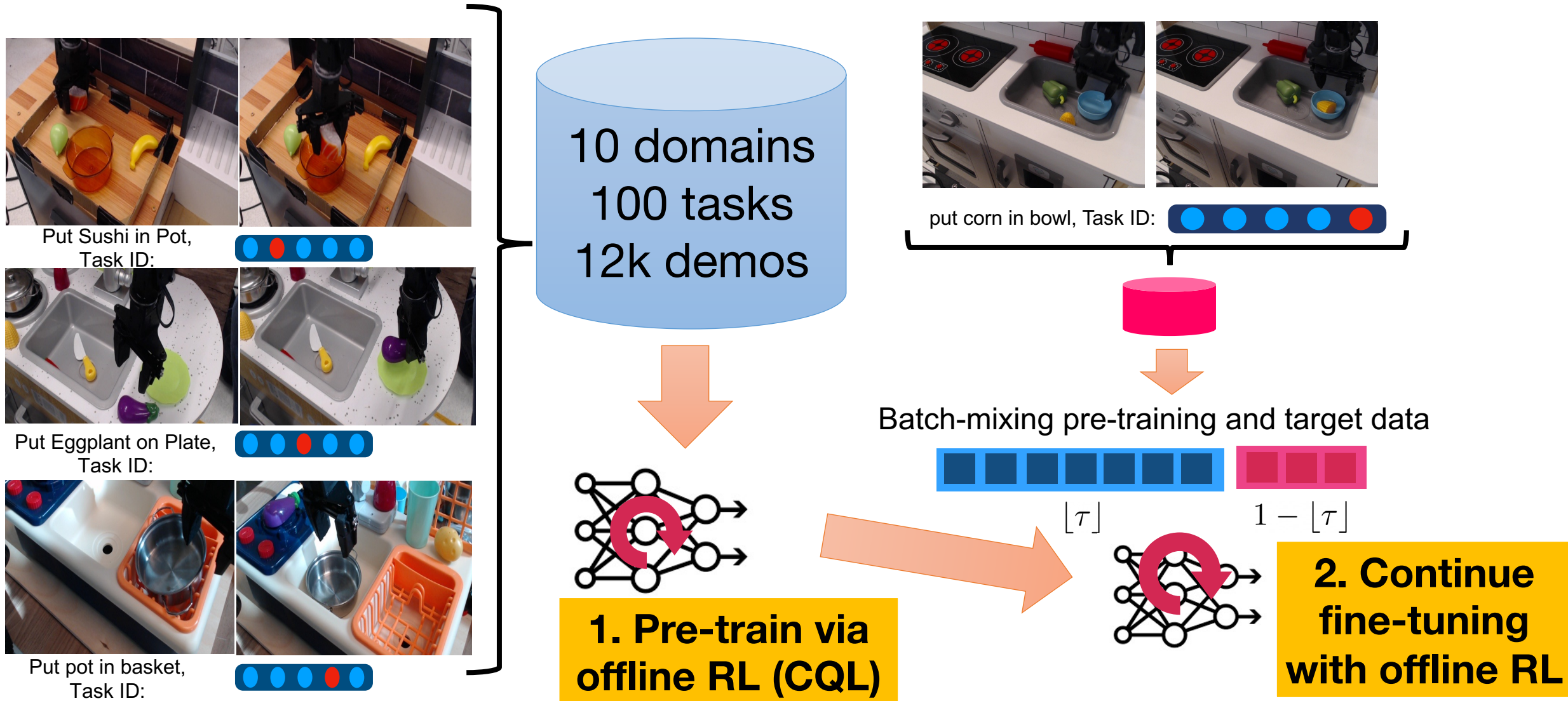
What We'll Cover in the Class

1. **Basics of RL:** definitions of MDP, POMDP, imitation vs RL
2. **RL algorithms:** model-free RL, model-based RL, value learning, policy gradients, hybrid algorithms, meta RL
3. **Exploration-exploitation tradeoff:** theory of exploration, bonus-based exploration, curiosity-driven exploration, unsupervised RL
4. **Offline RL & Scaling up RL:** pessimism, conservative value functions, policy constraints, model-based uncertainty estimation, connection to POMDPs
5. **RL for foundation models and foundation models for RL**
6. **Applications and real-world use cases**

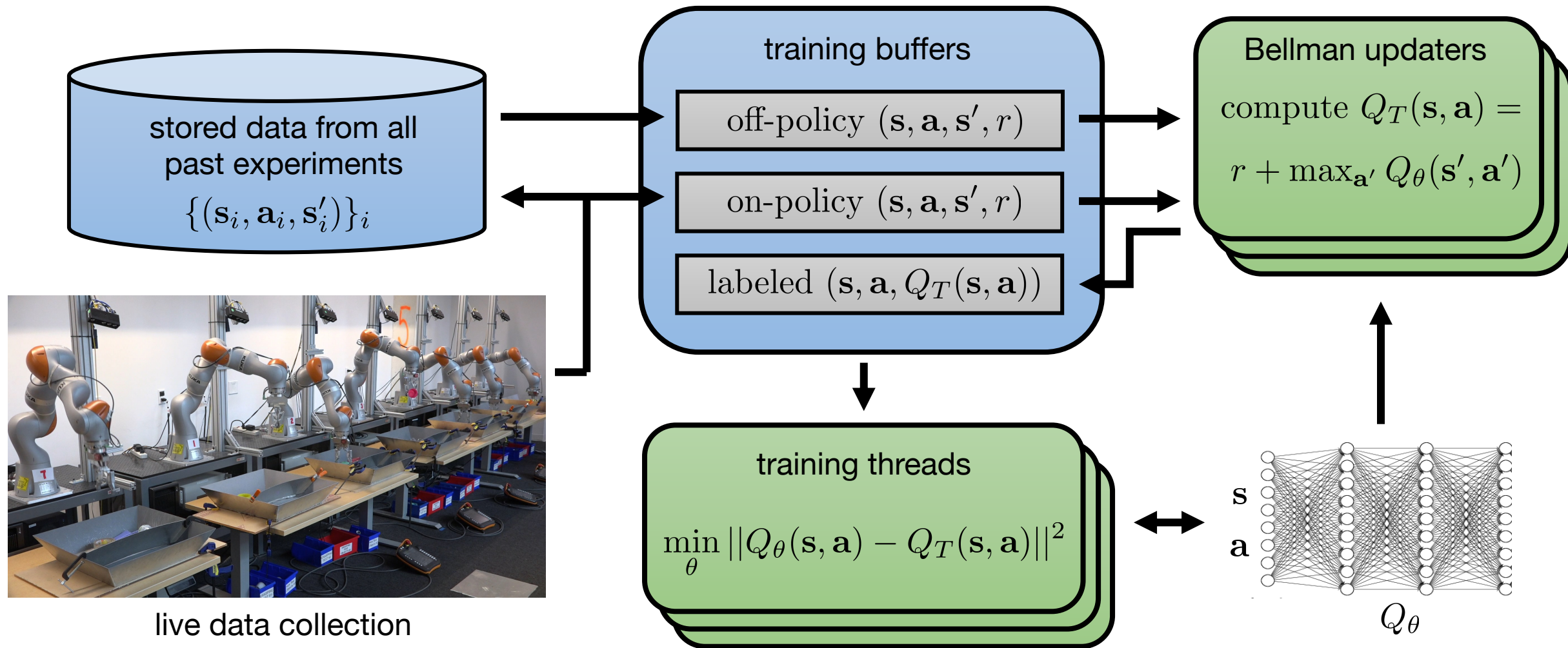
But you can do cool things with RL...

RL for Robotic Policy Learning

My Own: Pre-Training for Robots Using RL



Robot Learning





Kalashnikov, Irpan, Pastor, Ibarz, Herzog, Jang, Quillen, Holly, Kalakrishnan, Vanhoucke, Levine. **QT-Opt: Scalable Deep Reinforcement Learning of Vision-Based Robotic Manipulation Skills**

Achieving Human Level Competitive Robot Table Tennis

Table Tennis

David B. D'Ambrosio^{1,*}, Saminda Abeyruwan^{1,*}, Laura Graesser^{1,*}, Atil Iscen¹, Heni Ben Amor², Alex Bewley², Barney J. Reed^{2,†}, Krista Reymann², Leila Takayama^{2,§}, Yuval Tassa², Krzysztof Choromanski, Erwin Coumans, Deepali Jain, Navdeep Jaitly, Natasha Jaques, Satoshi Kataoka, Yuheng Kuang, Nevena Lazic, Reza Mahjourian, Sherry Moore, Kenneth Oslund, Anish Shankar, Vikas Sindhwani, Vincent Vanhoucke, Grace Vesom, Peng Xu, and Pannag R. Sanketi¹

via
Sim2Real

Google DeepMind

¹Primary contributors, ^{*}Corresponding authors (order randomized, equal contributions), ²Core contributors (Alphabetized),
[†]Work done at Google DeepMind via Stickman Skills Center LLC, [§]Work done at Google DeepMind via Hoku Labs.

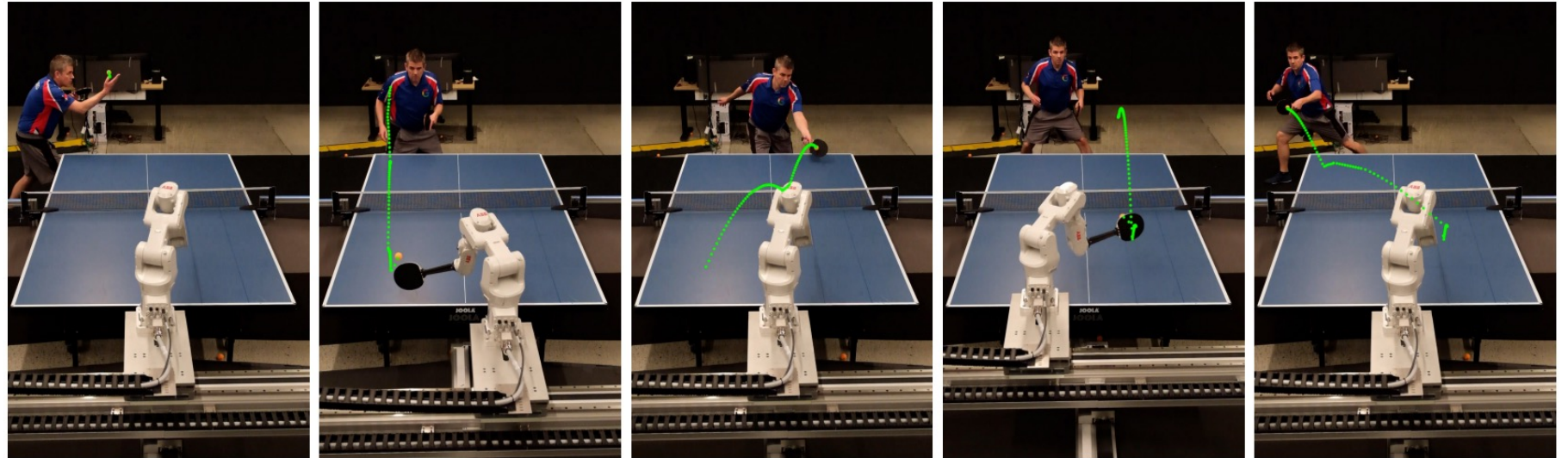
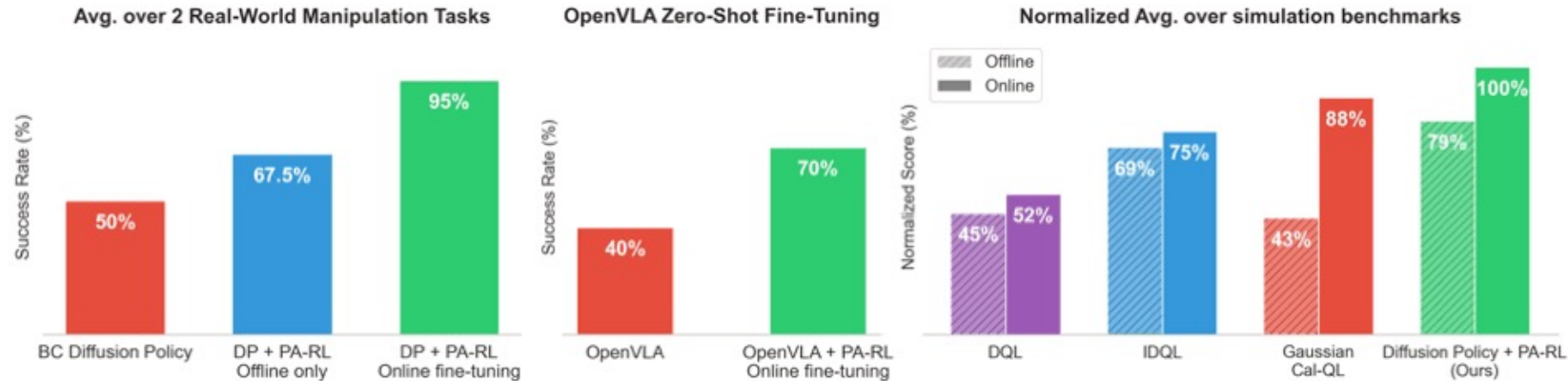
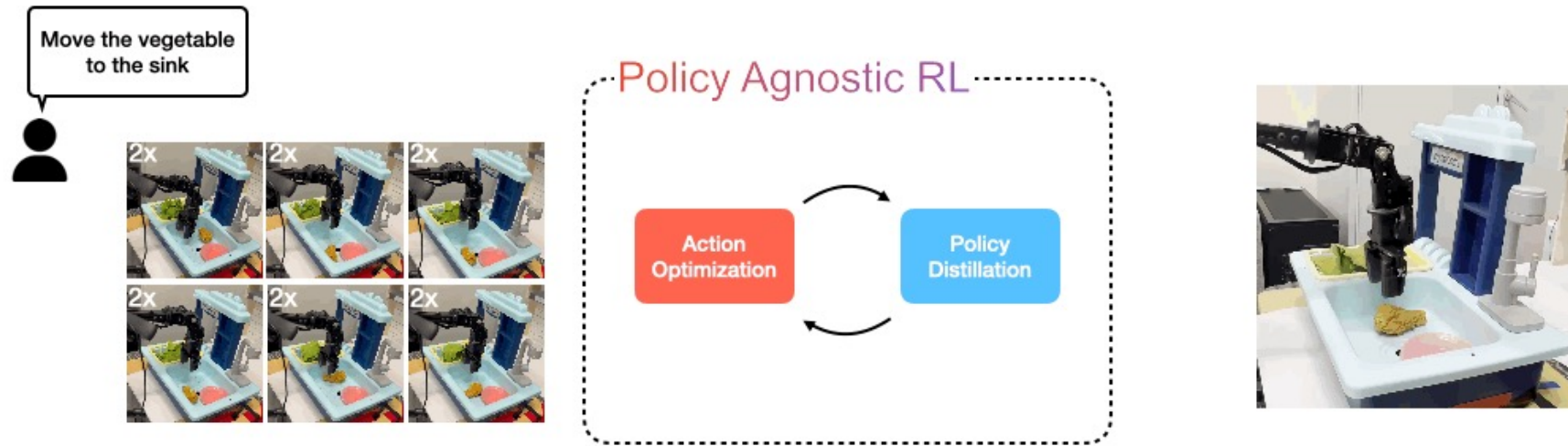


Fig. 1: Our table tennis robot playing against a professional coach. The green dots show the trajectory of the ball during the rally. The table tennis robot is a 6 DoF ABB 1100 arm mounted on top of two Festo linear gantries, enabling motion in the 2d plane. The x gantry, which moves side to side across the table, is 4m long and the y gantry, which moves towards and away from the table, is 2m long. A 3d printed paddle handle and paddle with short pips rubber is attached to the arm.

My Own: Fine-Tuning of Robot Policies



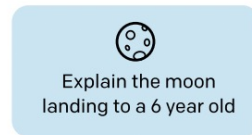
RL for Foundation Models

Aligning Large Language Models (e.g., ChatGPT)

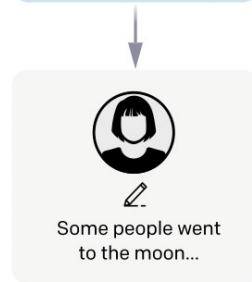
Step 1

Collect demonstration data, and train a supervised policy.

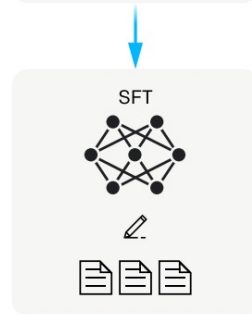
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



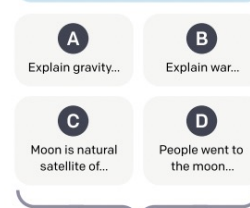
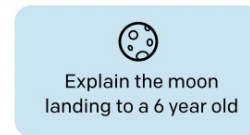
This data is used to fine-tune GPT-3 with supervised learning.



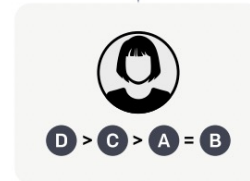
Step 2

Collect comparison data, and train a reward model.

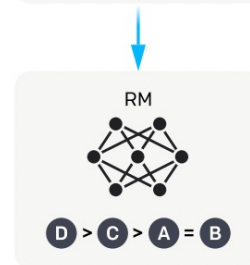
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



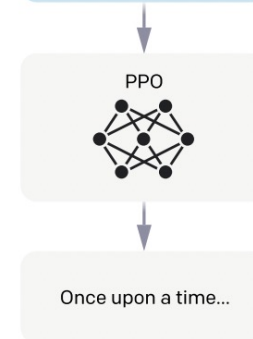
Step 3

Optimize a policy against the reward model using reinforcement learning.

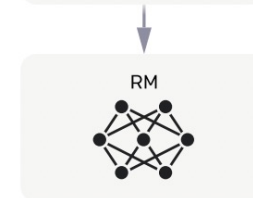
A new prompt is sampled from the dataset.



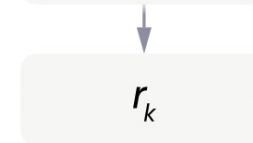
The policy generates an output.



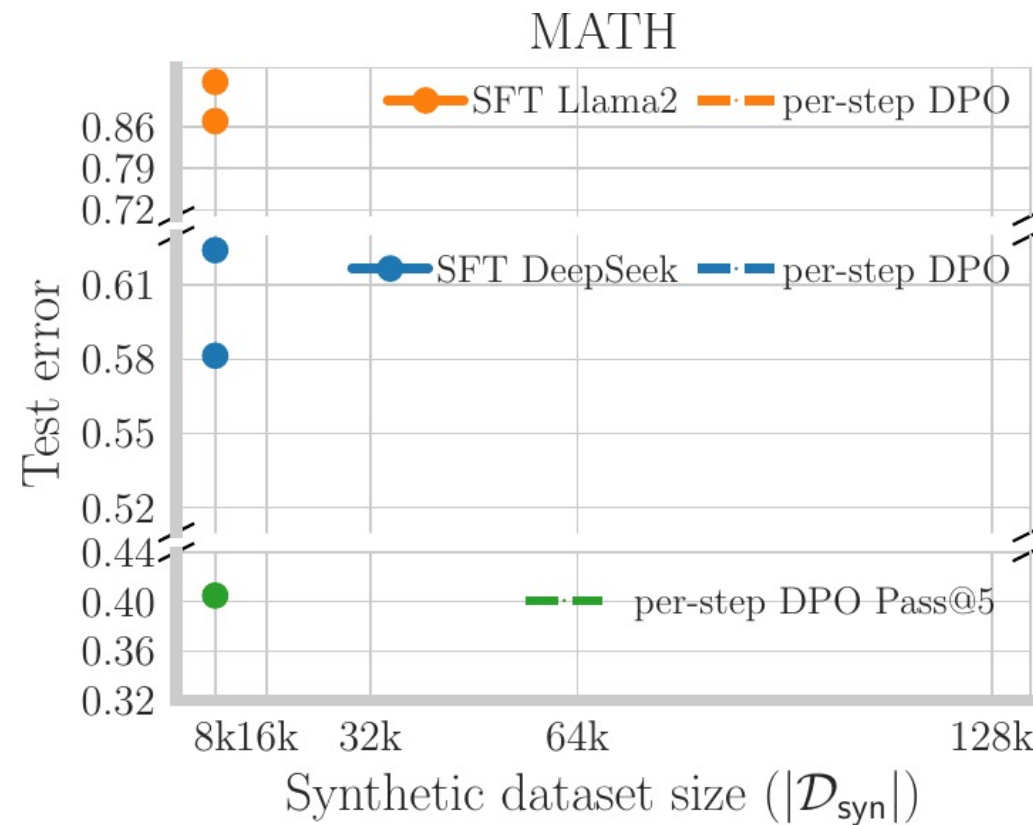
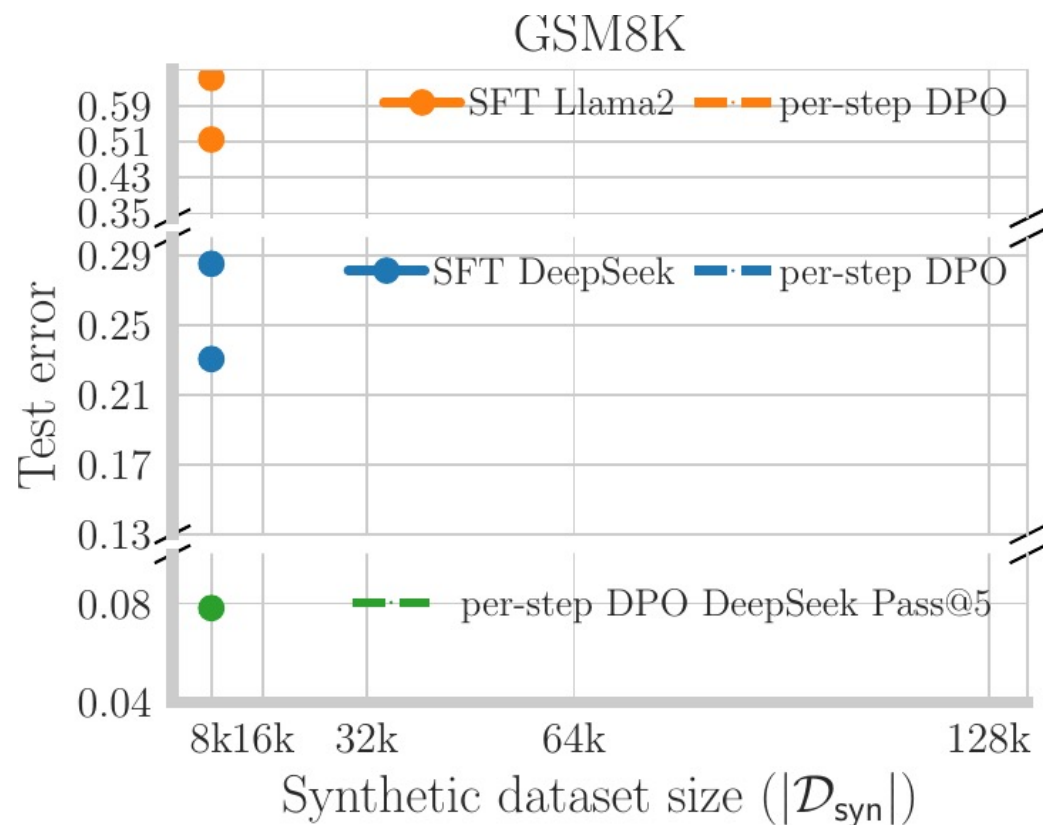
The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.

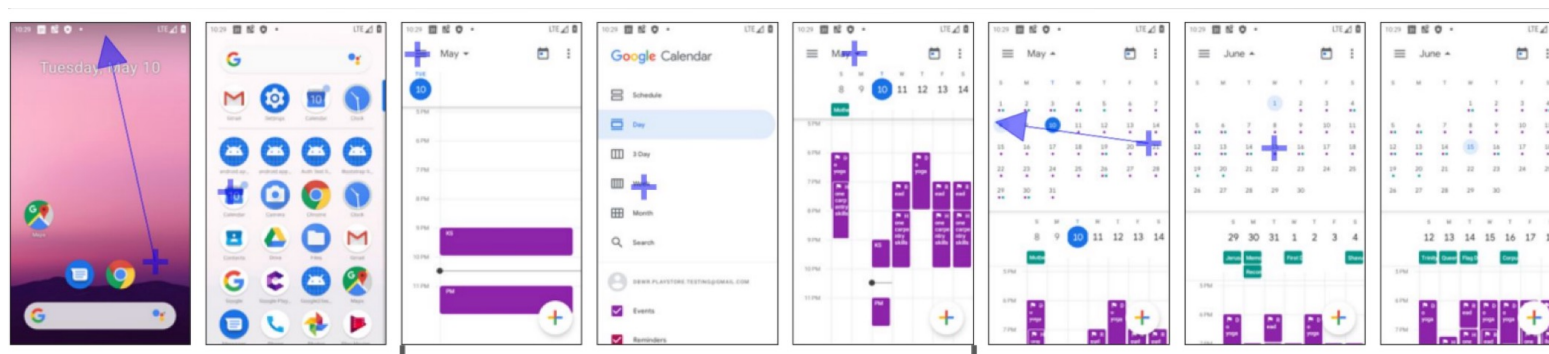
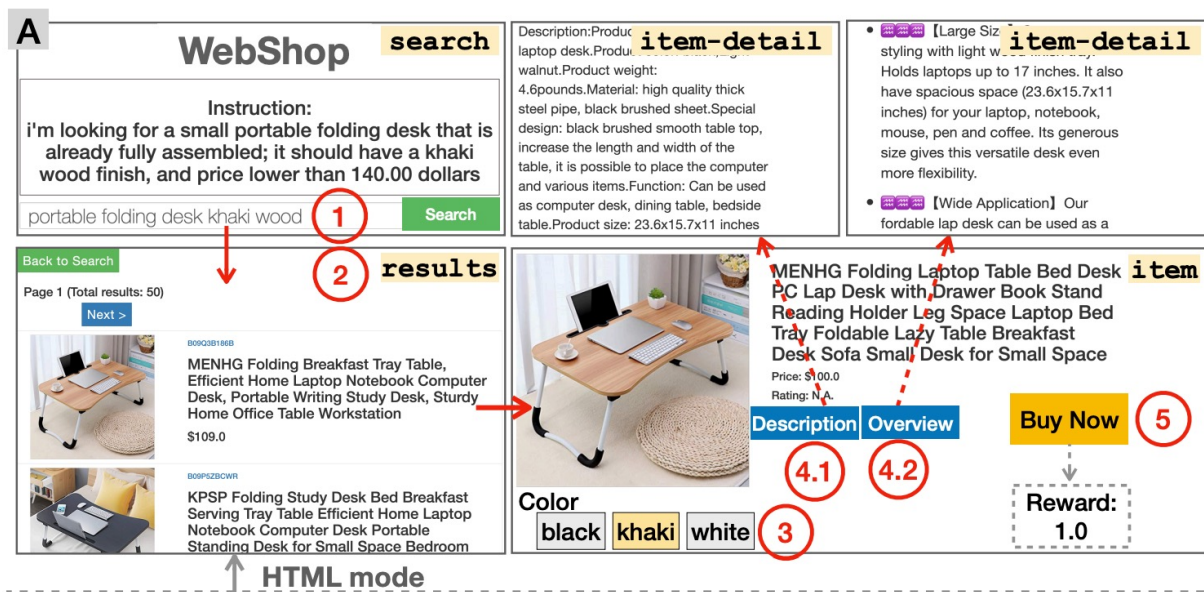
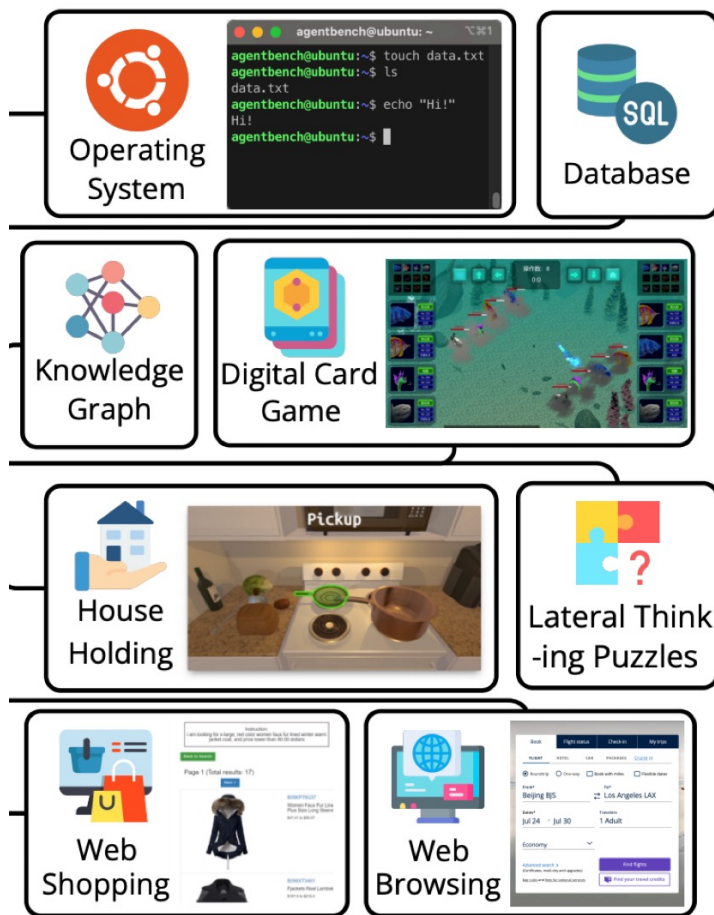


My Own: Improve Reasoning in LLMs by 8x



Finding: Using offline RL gives you an 8x data efficiency boost in performance over SFT.

My Own: RL for Training LLM and VLM Agents



Liu et al. **AgentBench: Evaluating LLMs as Agents.** arXiv 2024.

Yao et al. **WebShop: Towards Scalable Real-World Web Interaction with Grounded Language Agents.** NeurIPS 2022

Rawles et al. **Android in the Wild: A Large-Scale Dataset for Android Device Control.** 2023

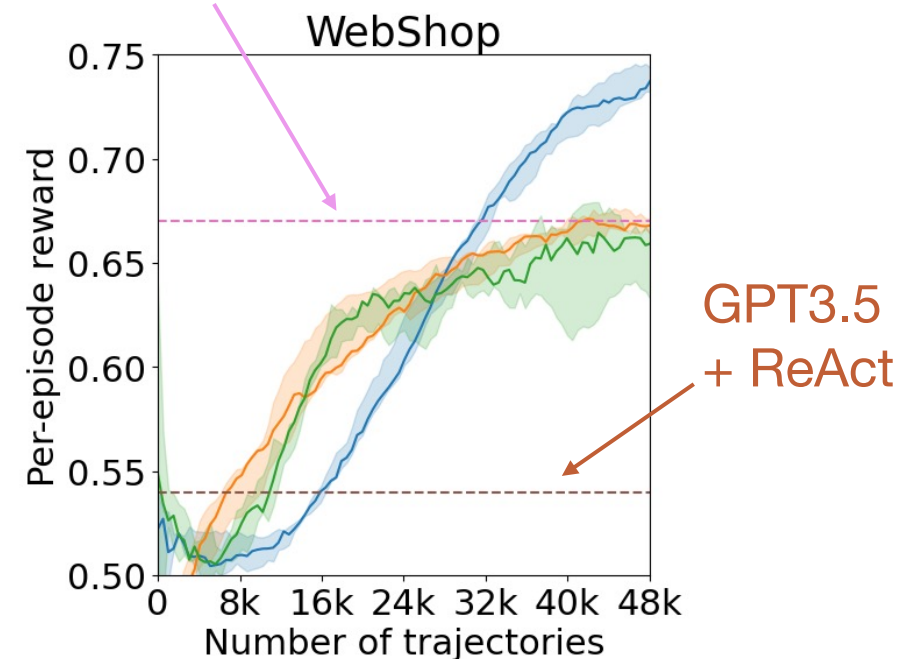
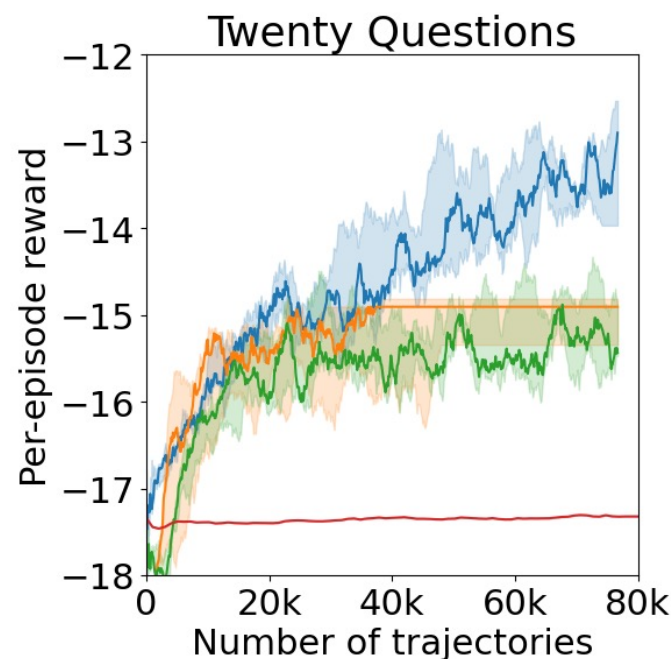
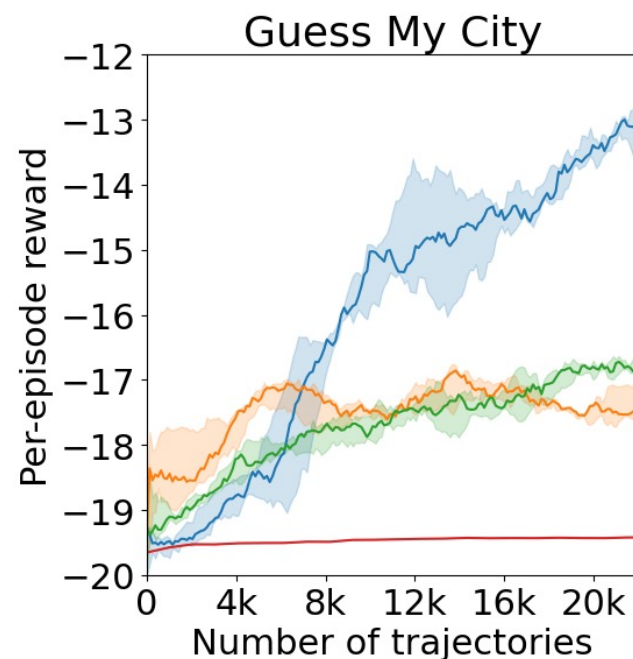
And To Be Precise, Multi-Turn RL

Results: Online Agent Tasks with GPT-2 models

Ask the right questions to identify what the hidden object (e.g., city, hidden word) is

Execute actions on the web

GPT3.5 with expert prompt



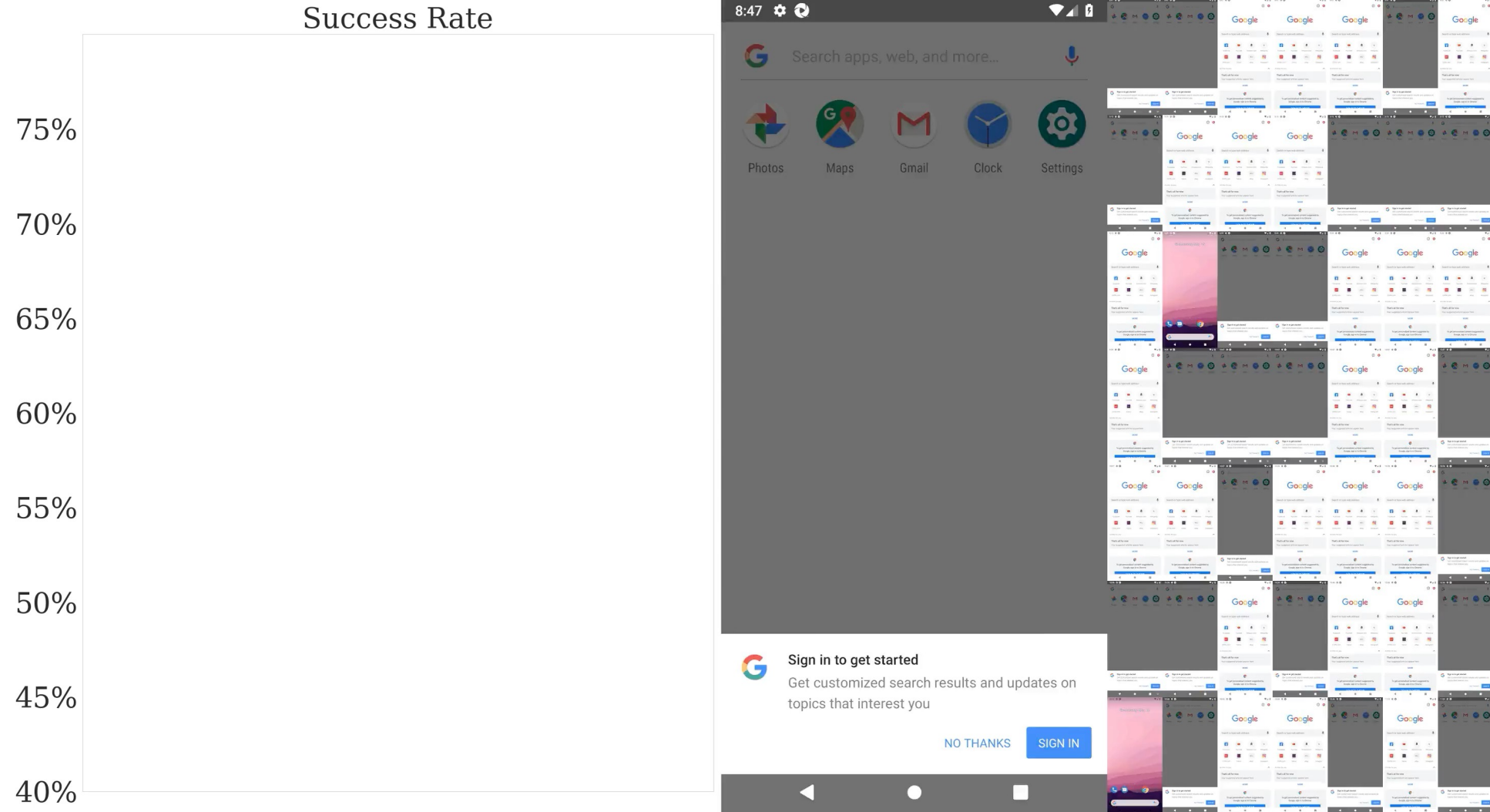
— ArCher — CHAI — Filtered BC - - - GPT3.5-expert-prompt - - - GPT3.5-ReAct

Zhou et al. **ArCher: Training Language Model Agents via Hierarchical Multi-Turn RL**. arXiv 2024.

Yao et al. **WebShop: Towards Scalable Real-World Web Interaction with Grounded Language Agents**. NeurIPS 2022.

Abdulhai et al. **LMRL Gym: Benchmarks for Multi-Turn Reinforcement Learning with Language Models**. arXiv 2023.

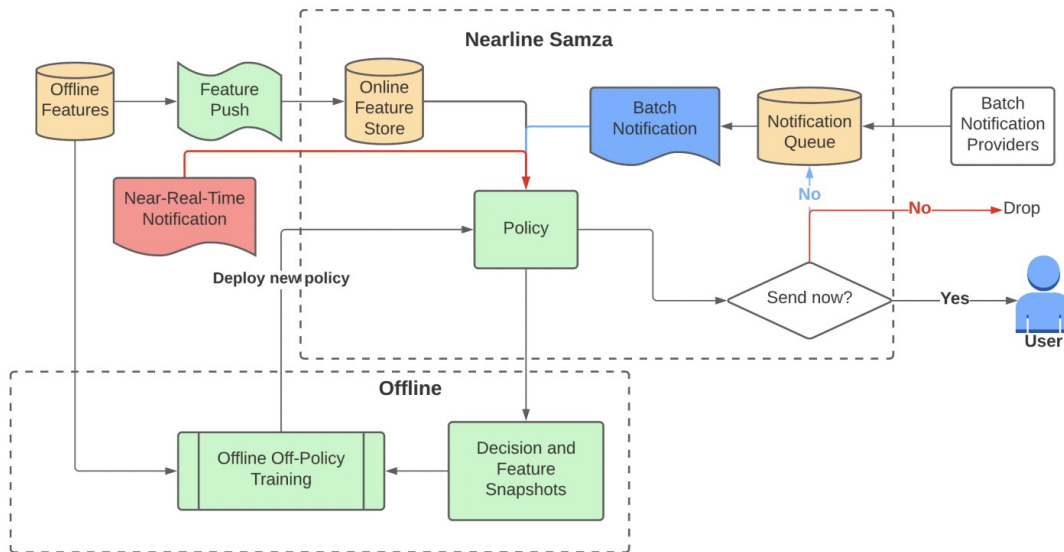
Autonomous Fine-Tuning: Device Control



RL for Systems and Other Applications

RL for Real-Time Notification Systems

Real-Time Mobile Notification Systems @ LinkedIn [Deployed]



Goal: Maximize user engagement by sending them meaningful notifications

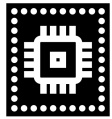
Metric	DDQN vs. Baseline	DDQN + CQL vs. Baseline
Sessions	not stat sig	+ 0.24%
WAU	-0.69%	+ 0.18%
Volume	+7.72%	-1.73%
CTR	-7.79%	+2.26%

fications. These two metrics are usually harder to move without increasing Volume than CTR metric, and hence +0.24% sessions and +0.18% WAU are considered significant business impact. We have ramped the DDQN + CQL model to all users based on this result.

My Own: RL for Accelerator Design

reward

$$\max_{\text{chip icon}} -\text{Runtime}_{\text{workload}}(\text{chip icon})$$



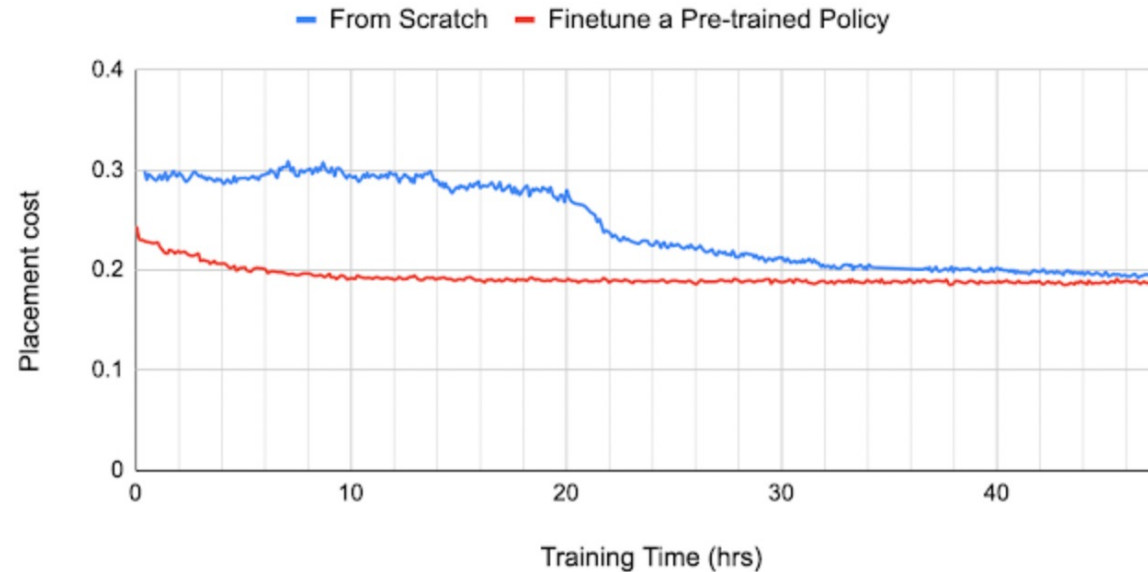
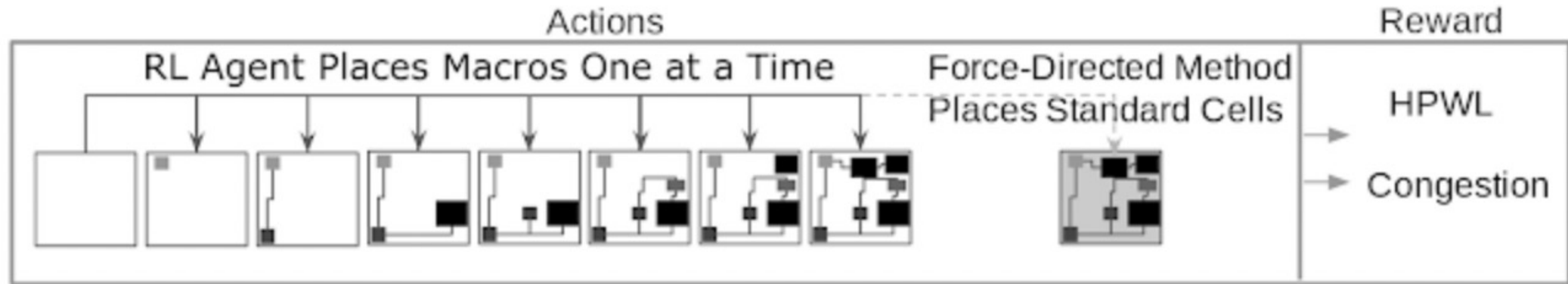
of Cores,
of Compute Lanes,
Core memory, ...

action

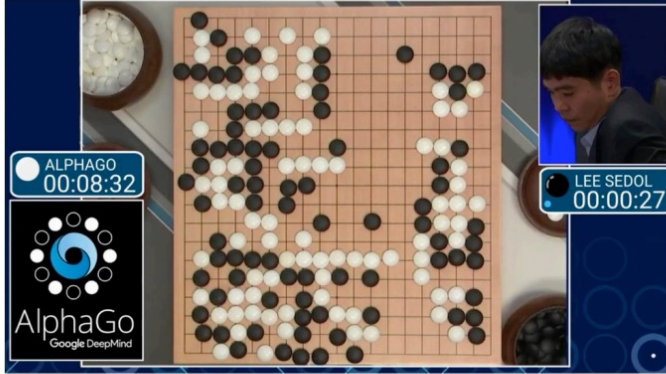
MobileNet,
U-Net,
ResNet 50

observation

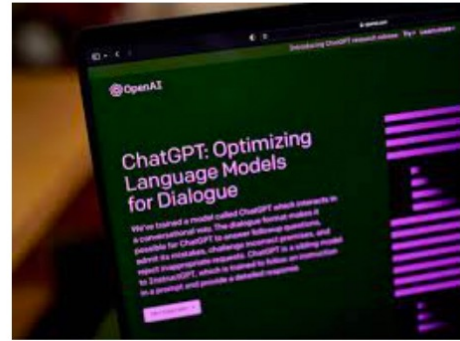
RL for Chip Placement



And many more applications....



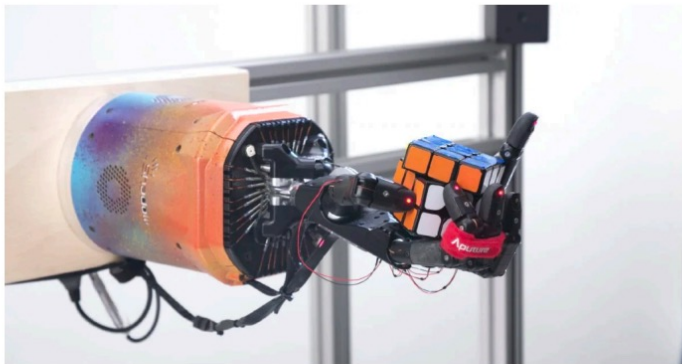
AlphaGo ('16)



ChatGPT ('22)



Stratospheric balloon navigation ('20)



Dexterous manipulation ('19)



Racing in Gran Turismo ('22)

But it is also not that straightforward / solved..

1. Lots of fundamental questions in RL:

- a) If you need to do better than data, you need to **discover** better decisions
- b) Exploration-exploitation tradeoff
- c) Learning to extrapolate / “on-the-tails” is not that easy
- d) Humans learn way faster than deep RL methods: how can we do it?

2. Lots of additional fundamental questions in deep RL:

- a) Deep networks work well for supervised learning, but do they for RL?
- b) How does RL scale up? Can it leverage advances from pre-training?
- c) How can deep RL use foundation models? And vice-versa?
- d) Can we integrate it into the paradigm of pre-train then fine-tune?
- e) How should practitioners tune RL methods?